

Reviewer Pack — Tropical Convex Hull Dimension and ACC^0 Circuit Size

Entry 9e0cac4cfcf9 · INCONCLUSIVE

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Tropical Convex Hull Dimension and ACC^0 Circuit Size

Entry ID: 9e0cac4cfcf9 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-07 20:41:09 UTC

1. Conjecture

Field A (mathematical branch): Tropical Convex Geometry **Field B** (complexity object): ACC^0 Circuit Size

Statement:

For any CNF formula Φ with n variables, the dimension of the tropical convex hull of its clause vectors is $\Theta(\log n)$ if and only if Φ has an ACC^0 circuit of size $\Theta(n)$.

Rationale (proposer’s reasoning):

Tropical convex hulls capture the extremal structure of clause interactions, which may constrain the algebraic complexity of ACC^0 circuits. The logarithmic scaling suggests a trade-off between geometric sparsity and computational efficiency.

Taxonomy category: ACC_LB_via_WILLIAMS (status at proposal time: partially_alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 73a63ba2c26a6ad3

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.95	SAFE	UNCERTAIN
NATURAL_PROOFS	SAFE	0.95	SAFE	UNCERTAIN
ALGEBRIZATION	SAFE	0.95	UNCERTAIN	SAFE
KARP_LIPTON	SAFE	0.95	SAFE	SAFE

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_3cnf(n):
        clauses = []
        for _ in range(2 * n):
            clause = [random.choice([-1, 1]) * (i + 1) for i in range(n)]
            if not any(clause[i] == -clause[j] for i in range(n) for j in range(i + 1, n)):
                clauses.append(clause)
        return clauses

    def tropical_convex_hull(clauses):
        # Placeholder for actual implementation
        return len(clauses)

    def acc0_circuit_size(clauses):
        # Placeholder for actual implementation
        return len(clauses) * 2

    n = random.randint(5, 40)
    clauses = generate_3cnf(n)
    hull_dimension = tropical_convex_hull(clauses)
    circuit_size = acc0_circuit_size(clauses)

    if hull_dimension == math.log(n, 2):
        conjecture_holds = True
        counterexample = ""
    else:
        conjecture_holds = False
        counterexample = "mapping_undefined"

    return {
        "metric_name": "hull_dimension",
        "metric_value": hull_dimension,
        "instances_tested": 1,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 1 for i in range(5, 30)]
```

```

results = []
for seed in seeds:
    result = run_trial(seed)
    print(f"TRIAL: {result}")
    results.append(result)

mean_value = sum(r["metric_value"] for r in results) / len(results)
std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(results))
support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

if all(r["conjecture_holds"] for r in results):
    print(f"RESULT: SUPPORTED mean={mean_value:.4f} std={std_value:.4f} support_fraction={support_fraction:.4f}")
elif any(not r["conjecture_holds"] for r in results):
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
    print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")
else:
    print("RESULT: INCONCLUSIVE mapping_undefined")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```

'conjecture_holds': False, 'counterexample': 'mapping_undefined'}
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 14, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 50, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 46, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 18, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 76, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 32, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 72, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 38, 'instances_tested': 1, 'conjecture_hold
TRIAL: {'metric_name': 'hull_dimension', 'metric_value': 34, 'instances_tested': 1, 'conjecture_hold

--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f6948cba.py", line 74, in <module>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~~
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_f6948cba.py", line 74, in <genexpr>
    first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
                        ~~~~~~
KeyError: 'seed'

```

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test crashed with KeyError 'seed' before producing valid data, preventing reliable assessment of conjecture status. | next: Fix test to handle 'seed' parameter properly and re-run with validated input configurations

11. Audit log (LLM calls)

Total LLM calls: 8

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	106068
2	propose	ollama_remote	qwen3:8b	0	0	103416
3	preregistration	ollama_remote	qwen3:8b	0	0	19810
4	novelty	ollama_remote	qwen3:8b	0	0	16542
5	novelty	ollama_remote	qwen3:8b	0	0	12585
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	19081
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7285
8	judge	ollama_remote	qwen3:8b	0	0	12543

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 297332 ms total latency. Provider mix: {'ollama_remote': 8}

(full prompt+response transcripts available in `research/audit/9e0cac4cfcf9.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/9e0cac4cfcf9.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/9e0cac4cfcf9.tar.gz` (if generated)